

EzAssist: AI-Powered Knowledge Assistant for Faster Support Issue Resolution

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Abstract

EzAssist is an AI-powered Knowledge Assistant that transforms how HPE Ezmeral support engineers find and apply troubleshooting information. Instead of manually searching across multiple Knowledge Base (KB) articles, PDF documents, and Salesforce records, engineers can ask a question in natural language and receive consolidated, step-by-step troubleshooting guidance from a single interface. The system uses a Retrieval-Augmented Generation (RAG) architecture combining a locally-hosted LLaMA 2 7B large language model with semantic search over PostgreSQL+pgvector to deliver context-aware answers grounded exclusively in approved support documentation. Engineers can also upload new PDF-based KB articles and troubleshooting guides, ensuring the knowledge base grows continuously. A working Proof of Concept is deployed and validated on an HPE Ezmeral Kubernetes cluster.

Problem Statement

Support Engineers spend a significant amount of time searching across multiple Knowledge Base (KB) articles, support documents, troubleshooting guides, release notes, and internal repositories to diagnose and resolve customer issues. This creates several challenges:

1. Knowledge is Distributed Across Multiple Sources

Critical troubleshooting information is scattered across KB articles, PDF documents, product manuals, support notes, and internal documentation. Engineers often need to search multiple systems before finding the required information.

2. Increased Time to Resolution

A considerable portion of troubleshooting time is spent locating relevant documentation rather than solving the customer's problem. This increases Mean Time to Resolution (MTTR) and directly impacts customer satisfaction.

3. Knowledge Discovery Becomes Difficult

As the volume of support content grows, identifying the most relevant article or procedure becomes increasingly challenging, especially for new engineers or when handling unfamiliar product areas.

4. Repeated Effort Across Support Teams

Different engineers frequently spend time searching for the same information and recreating troubleshooting steps that already exist in documented knowledge sources, leading to duplicated effort across global teams.

5. Limited Ability to Leverage New Knowledge

Newly created KB articles, field advisories, and troubleshooting guides are difficult to consolidate into a single searchable knowledge repository that all engineers can benefit from.

Our Solution

We developed EzAssist, a Hybrid RAG Knowledge Assistant that serves as a single-stop intelligent support companion for HPE Ezmeral Support teams. Instead of manually searching through hundreds of KB articles and documents, engineers simply ask a question in natural language and receive:

- Relevant troubleshooting steps extracted from matching KB articles

- Associated commands and procedures required for issue resolution
- Consolidated information from multiple knowledge sources (PDFs + Salesforce)
- Context-aware answers specific to the HPE Ezmeral product line
- Continuous knowledge expansion through PDF document upload

Technical Architecture

LLM Engine — LLaMA 2 7B Chat (Q4_K_M quantized, ~4 GB GGUF): Runs entirely within the Kubernetes pod via llama-cpp-python with a 4,096-token context window, temperature=0.2 for deterministic output, top_p=0.9, and repeat_penalty=1.2. No external API calls — all data stays within the cluster, addressing data sovereignty requirements.

Embedding Model — all-MiniLM-L6-v2 (Sentence-Transformers): Generates 384-dimensional dense vector embeddings for document chunks and user queries. Enables semantic similarity matching that goes beyond keyword search to understand the meaning of support queries.

Vector Database — PostgreSQL 15 + pgvector: Stores document chunk embeddings and performs L2 distance-based nearest-neighbor search. Two-pass retrieval: first finds top-k closest chunks globally, then expands within the same source document to capture complete procedures. Auto-detects document style (numbered steps vs. KB-style Issue/Cause/Resolution) to optimize chunk ordering.

PDF Ingestion Pipeline: Multi-stage processing -> text extraction (pdfminer) -> unicode normalization -> line deduplication -> hyphenation repair -> n-gram repeat collapsing -> section-aware chunking (preserving headers and code blocks) -> overlapping window assembly -> sentence-transformer embedding -> pgvector storage. CLI commands (kubect!, systemctl, helm, bdconfig) are preserved exactly as documented.

Salesforce Knowledge Integration: Dual-method search using SOSL (full-text) and SOQL (LIKE-based fallback) against Knowledge Articles with product-line filtering. Rich-text HTML bodies are converted to clean text using html2text. Supports both Session ID and OAuth2 authentication.

Answer Generation: Retrieved context chunks are cleaned and assembled into a prompt with strict grounding rules, the model uses only commands found exactly in the retrieved context and refuses to answer when context is irrelevant, preventing fabricated troubleshooting procedures.

Deployment: Fully containerized with Docker, deployed on Kubernetes using Helm charts with init containers, Gunicorn WSGI server with --preload, and an automated deployment script for rapid cluster provisioning.

Evidence the Solution Works

The system is deployed as a working POC on an HPE Ezmeral Kubernetes cluster (K8s v1.30.14, Rocky Linux 8.10) and validated with real support scenarios:

Scenario 1 — Correct Procedure Retrieval

Query: “What are the steps for manual restart ERE?” — The system correctly retrieved the 7-step restart procedure from the uploaded KB document including exact systemctl commands (bds-monitoring, bds-worker, bds-controller) and the required suspendha pre-step.

Scenario 2 — Answer Reliability

When queried about topics not covered in the knowledge base, the system responds with “I don’t have enough information in the knowledge base to answer this question” instead of generating plausible-sounding but incorrect procedures — a critical safety requirement for production support.

Scenario 3 — Product-Line Filtering

Active filtering ensures only relevant HPE Ezmeral Container Platform articles are returned, excluding articles from other product families (Data Fabric, Unified Analytics).

Competitive Approaches

Approach	Advantages	Disadvantages
Generic Cloud LLMs (ChatGPT, Copilot)	High quality generation; easy to use	Data leaves HPE network; no SFDC integration; hallucination risk
HPE GreenLake AI / PCAI	Enterprise-grade GPU inference	Requires GPU infrastructure; not targeted at support workflows
Manual Salesforce Search	Accurate when engineers know what to look for	Slow; no semantic search; no cross-source correlation
Generic RAG Frameworks (LangChain)	Flexible development framework	Single-source only; no SFDC integration; no product-line filtering

EzAssist uniquely combines: natural language search across PDFs and live Salesforce Knowledge Articles, fully offline LLM with no data exfiltration, answers grounded exclusively in approved documentation, continuous knowledge expansion through PDF upload, and Helm-packaged deployment for any Kubernetes cluster.

Current Status

The solution is a fully functional Proof of Concept deployed on an HPE internal Kubernetes cluster. It is actively used for HPE Ezmeral Runtime Enterprise support knowledge retrieval. The Docker image is published (sabya610/rag-app:v2), the Helm chart is packaged (rag-app-0.2.0.tgz), and the automated deployment script has been validated for rapid provisioning on new clusters. The web interface provides three views: Q&A (with PDF/SFDC/Both source selection), PDF Upload, and Query History.

Next Steps

1. GPU-Accelerated Inference: Integrate with HPE Ezmeral PCAI/KServe for GPU-based LLM inference to reduce response times.
2. Automated Salesforce Authentication: Replace manual Session ID management with OAuth2 Connected App flow for unattended operation.
3. User Feedback Loop: Add answer rating (thumbs up/down) to improve retrieval ranking over time.
4. Case History Integration: Ingest case comments and resolution notes from Salesforce Cases, enabling learning from historical resolutions.
5. Multi-Product Support: Parameterize product-line filtering to support other HPE product families (GreenLake, Aruba, Storage).

References

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